

Cybersecurity Basics

An Introduction to Core Concepts, Threats, and Attack Vectors

Introduction

Cybersecurity refers to the practice of protecting computer systems, networks, applications, and data from unauthorized access, damage, or disruption. As organizations and individuals rely more heavily on digital systems for communication, storage, and transactions, understanding the fundamental principles of cybersecurity has become essential. This assignment covers the foundational concept of the CIA Triad, common types of cyber threats, and the various attack vectors through which these threats are carried out.

1. The CIA Triad

The CIA Triad is a foundational model used to guide policies and evaluate the security posture of an organization. It consists of three core principles that every secure system must aim to uphold.

Confidentiality

Confidentiality ensures that sensitive information is accessible only to those who are authorized to view it. It prevents unauthorized individuals from gaining access to private data such as personal records, financial details, or trade secrets. Techniques used to maintain confidentiality include encryption, access control lists, authentication mechanisms, and data classification policies.

Integrity

Integrity refers to maintaining the accuracy, consistency, and trustworthiness of data over its entire lifecycle. It ensures that information is not altered or tampered with, whether accidentally or maliciously, during storage or transmission. Methods such as checksums, hashing algorithms, digital signatures, and version control help preserve data integrity.

Availability

Availability guarantees that information and resources are accessible to authorized users whenever they are needed. It focuses on ensuring that systems remain operational and are not disrupted by hardware failures, software issues, or malicious attacks such as Denial of Service. Redundancy, regular maintenance, backups, and failover systems all contribute to maintaining availability.

2. Threat Types

A cyber threat is any potential danger that could exploit a vulnerability to breach security and cause harm. Below are some of the most common types of threats encountered in the digital world.

Phishing

Phishing is a deceptive practice in which attackers impersonate a trustworthy entity, usually through email, text messages, or fake websites, in order to trick victims into revealing sensitive information such as passwords, credit card numbers, or login credentials. Phishing often relies on urgency or fear to manipulate the victim into acting quickly without verifying the source.

Malware

Malware, short for malicious software, is any program or code specifically designed to infiltrate, damage, or gain unauthorized access to a computer system without the consent of the user. It includes viruses, worms, trojans, spyware, and adware, each with different methods of infection and objectives.

DDoS (Distributed Denial of Service)

A DDoS attack occurs when multiple compromised systems, often forming a botnet, are used to flood a target server, service, or network with excessive traffic. This overwhelms the target's resources and makes it unavailable to legitimate users, directly attacking the availability principle of the CIA Triad.

SQL Injection

SQL Injection is a code injection technique in which an attacker inserts malicious SQL statements into an input field or query, exploiting vulnerabilities in a web application's database layer. If successful, it can allow the attacker to view, modify, or delete sensitive data stored in the database, or even gain administrative control over it.

Brute Force

A brute force attack is a trial-and-error method used to guess login credentials, encryption keys, or hidden information by systematically trying every possible combination until the correct one is found. Although time-consuming, this method can be effective against weak or short passwords, especially when automated tools are used.

Ransomware

Ransomware is a type of malicious software that encrypts a victim's files or locks them out of their own system, demanding payment, usually in cryptocurrency, in exchange for restoring access. It is one of the most financially damaging forms of cyberattack, often targeting businesses, hospitals, and government institutions.

3. Attack Vectors

An attack vector is the specific path or method that an attacker uses to gain unauthorized access to a system or network in order to deliver a malicious outcome.

Social Engineering

Social engineering is a manipulation technique that exploits human psychology rather than technical vulnerabilities to trick individuals into divulging confidential information or performing actions that

compromise security. This can include tactics like pretexting, baiting, or impersonation, and it often serves as the entry point for larger attacks such as phishing.

Wireless Attacks

Wireless attacks target vulnerabilities in wireless networks, such as Wi-Fi, to intercept data, gain unauthorized network access, or disrupt communication. Common examples include eavesdropping on unsecured networks, setting up rogue access points, and exploiting weak encryption protocols to capture sensitive information transmitted over the air.

Insider Threats

An insider threat originates from within the organization itself, typically involving current or former employees, contractors, or business partners who have legitimate access to systems and data. These threats can be intentional, such as data theft for personal gain, or unintentional, resulting from negligence or lack of awareness, but both can cause significant harm to an organization's security.

Conclusion

Understanding the CIA Triad, common threat types, and typical attack vectors forms the foundation of cybersecurity knowledge. By recognizing how confidentiality, integrity, and availability can be compromised, and by being aware of the methods attackers use, individuals and organizations can take proactive steps to defend their systems, safeguard sensitive data, and build a more resilient digital environment.